Leo Wang

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Skills

Languages ◇ **Advanced**: C#, Python
Intermediate: TypeScript/JavaScript, HTML/CSS, Java, C++
Beginner: HLSL, Matlab

Technologies ◇ **Platforms**: Unity, AWS, NextJS
Libraries: Dear ImGUI, Newtonsoft, OpenCV, Numpy, Selenium, Tensorflow, Matplotlib, TailwindCSS, React, Vite
Dev. Tools: Visual Studio, Android Studio, Visual Studio Code, Git, Github

Skills ◇ Software Optimization, Large-Scale Data Processing, Data Visualization, Web Development

Education

2024-26 ◇ **University of Maryland, Bachelor of Science**.
├ Double Major: **Computer Science** (Data Science) and **Math**.
├ GPA: 4.0
└ Was accepted into the Advanced Cybersecurity Experience for Students (ACES) Honors College.

2024 ◇ **UMD Presidential Scholarship** ($5,000/year)

2024 ◇ **National Merit Scholarship** ($1,000/year)

Experiences

2024 ◇ **Cybersecurity Training Platform for MITRE Corporation**. Web Developer, UMD
└ I worked with a student led team to develop a user-friendly yet powerful cybersecurity training platform for new MITRE employees. I built the frontend section of the admin-side pages using NextJS, TailwindCSS and Typescript.

2023 ◇ **IEEE Aerospace Conference**. Co-Author, Virtual
├ **"Enhancing Space Communications: A Novel Approach to Solving the Multi-Satellite Scheduling Problem"**
└ Published paper on work done during the NASA internship to the 2024 IEEE Aerospace Conference.

2023 ◇ **IEEE Integrated STEM Education Conference**. Presenter, Johns Hopkins Applied Physics Lab
├ **"Enhancing STEM Education to Communities with Low Access to STEM"**
└ Presented at IEEE Integrated STEM Education Conference on raising STEM engagement in under-served communities.

2021-23 ◇ **National Aeronautics and Space Administration (NASA)**. Intern, Goddard Space Flight Center
└ I led a team of 5 interns as lead developer to create a high fidelity and performant planetary terrain/orbiting satellite simulation that outperformed existing tooling in C# using the Unity Engine. (See Projects)

Projects

2024 ◇ **Personal Website**.
└ Designed and implemented a fully custom personal website using React, TailwindCSS, and Vite.

2024 ◇ **N-Body (Barnes-Hut) Simulation**.
└ Developed an optimized N-body simulation (**n=15,000**) by leveraging the Barnes-Hut algorithm in C++ and Dear ImGUI.

2022-23 ◇ **NASA Terrain System**.
├ I created an extremely powerful planetary terrain system that was the backbone of my NASA internship, capable of dynamically loading terrain data at arbitrary resolutions in realtime.
├ The internal database stores highly compressed height maps ranging from **tens of millions** to **billions of points**. (ex. one region stores 170 million points in 20 mib).
└ The system can process, render, and simulate up to **13,000,000 points per second** (including database read times).